

Unit 2: Regular Language and Finite Automata

Find a regular expression corresponding to each of the following subset of $\{a, b\}^*$:

- The set of all the string with length 5 exactly and next to last symbol is 'b'
- The set of all strings that ends with sub string 'abb'
- The set of all strings having 'a' as the first symbol and 'b' as the last symbol.

- ⇒
- $[(a+ab)(a+ab)(a+ab)b(a+ab)]$
 - $[(a+ab)^*abb]$
 - $a(a+ab)^*b$

Q2) Write regular expressions for the following.

- Binary numbers that are multiple of 2.
- Strings of a's and b's with no consecutive a's.
- Strings of a's and b's.

⇒

i) ~~10, 100, 110~~

→ $[(0+1)^*0]$

ii)

$(b+ab)^*(\epsilon+a)$

iii) $(a+ab)^*aa(a+ab)^*$

Q3) Find a regular expression of following subset of $\{0, 1\}^*$

- The language of all strings that begin or end with 00 or 11.
- The language of all strings ending with 1 and not containing